

2.12] $\int_0^\infty e^{-ax} \ln(x) dx$

* look at f^n at both limits:

(i) $x \rightarrow 0$: $\lim_{x \rightarrow 0} e^{-ax} \ln(x) = 1 \lim_{x \rightarrow 0} \ln(x)$

$$\sim \int_0^\infty \ln(x) dx = (x \ln(x) - x) \Big|_0^\infty = () \Big|_\infty - \lim_{x \rightarrow 0} (x \ln(x) - x) = () \Big|_\infty - 0$$

$a > 0$: $0 < e^{-ax} \leq 1$

* $|e^{-ax} \ln(x)| \leq |\ln(x)|$ *

$$\int_0^\infty |\ln(x)| dx \Rightarrow \int_0^\infty |e^{-ax} \ln(x)| dx \leq \int_0^\infty |\ln(x)| dx$$

only looking $x \rightarrow 0$: converge in the limit $x \rightarrow 0$

* Two ways $\lim_{n \rightarrow \infty} \int_0^n e^{-ax} \ln(x) dx$

$$\text{or } \int_0^\infty e^{-ax} \ln(x) dx = \underbrace{\int_0^b e^{-ax} \ln(x) dx}_* + \int_b^\infty e^{-ax} \ln(x) dx$$

(ii) look $x \rightarrow \infty$: $\lim_{x \rightarrow \infty} \left(\frac{\ln(x)}{e^{ax}} \right) = 0$ *

$|e^{-ax} \ln(x)| \leq |\ln(x)|$: can't use

$\int \ln(x) dx = x \ln(x) - x$: diverges $x \rightarrow \infty$

• Need to Bound $e^{-ax} \ln(x)$

$\lim_{x \rightarrow \infty} \ln(x) < e^{ax}$

$e^{-ax} \ln(x) < e^{-ax/2} \cdot e^{-ax/2}$

$e^{-ax} \ln(x) < e^{-\frac{ax}{2}}$

$\lim_{x \rightarrow \infty} \left(e^{-\frac{ax}{2}} \right) = 0$

$$\begin{aligned} * \int_b^\infty |e^{-ax} \ln(x)| dx &\leq \int_b^\infty |e^{-ax/2}| dx = -\frac{2}{a} e^{-ax/2} \Big|_b^\infty \\ &= \left(-\frac{2}{a} \right) \left(-e^{-ab/2} \right) \\ &= \frac{2}{a} e^{-ab/2} \end{aligned}$$

• Convergent in the limit $x \rightarrow \infty$

* To do this $\int_0^\infty e^{-ax} \ln(x) dx$ use Gamma fn $\Gamma'(x)$

$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$ $t \in \mathbb{R}$

• $\Gamma(n) = (n-1)! = \int_0^\infty x^{n-1} e^{-x} dx$ $n \in \mathbb{Z}^+$

$$\frac{d}{dt}(x^{t-1}) = \frac{d}{dt}(e^{\ln(x^{t-1})}) = \frac{d}{dt}(e^{(t-1)\ln(x)}) = \ln(x) x^{t-1}$$

$$\frac{d}{dt} \Gamma(t) = \int_0^\infty \ln(x) x^{t-1} e^{-x} dx = \Gamma'(t)$$

$$\Gamma'(1) = \int_0^\infty \ln(x) e^{-x} dx$$

i) $\Gamma'(1) = -\gamma = 0.57721$ Euler - Mascheroni constant.

$$-\gamma = \int_0^\infty \ln(x) e^{-x} dx$$

$$1/y = \frac{x}{a} \Rightarrow x = ay$$

$$e^{-ay}$$

$$-\gamma = \int_0^\infty \ln(ay) e^{-ay} (a dy) = a \int_0^\infty (\ln(a) + \ln(y)) e^{-ay} dy$$

$$= a \int_0^\infty e^{-ay} \ln(a) dy + a \int_0^\infty \ln(y) e^{-ay} dy$$

$$-\gamma = a \ln(a) \int_0^\infty e^{-ay} dy + a I$$

$$\Gamma(x+1) = x \Gamma(x)$$

$$\frac{1}{a} (-\gamma - a \ln(a) \int_0^\infty e^{-ay} dy) = I$$

$$\frac{1}{a} (-\gamma - a \ln(a) (\frac{2}{a})) = I$$

$$* \Gamma'(2)$$

2.13] [1] $f: [0, \infty) \rightarrow \mathbb{R}$

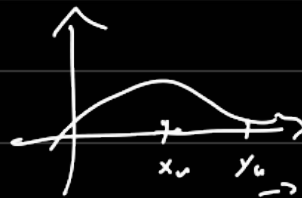
$$\int_0^\infty f(t) dt :$$

$$(x_n)_n, (y_n)_n \rightarrow \infty$$

$$\lim_{n \rightarrow \infty} \int_{x_n}^{y_n} f(t) dt = 0$$

$$* F(x) = \int_a^x f(t) dt$$

$$[x_n, y_n] \xrightarrow{n \rightarrow \infty}$$



$$* F(x) = \int_0^x f(t) dt$$

$$\int_{x_n}^{y_n} f(t) dt = \int_{x_n}^0 f(t) dt + \int_0^{y_n} f(t) dt$$

ex: $\int_1^2 f(x) dx = \int_1^0 f(x) dx + \int_0^2 f(x) dx$

$$= - \int_0^{x_n} f(t) dt + \int_0^{y_n} f(t) dt$$

$$\int_{x_n}^{y_n} f(t) dt = -F(x_n) + F(y_n)$$

$$* F(x) = \int_0^x f(t) dt$$

* $\lim_{x \rightarrow \infty} F(x) = \int_0^\infty f(t) dt$: finite i.e. also converges $\Rightarrow$ some number

$$\lim_{x \rightarrow \infty} F(x) = \int_0^\infty f(t) dt = N \in \mathbb{R}$$

$$\lim_{n \rightarrow \infty} \int_{x_n}^{y_n} f(t) dt = \lim_{n \rightarrow \infty} (F(y_n) - F(x_n))$$

$$\lim_{n \rightarrow \infty} F(y_n) = \lim_{n \rightarrow \infty} \int_0^{y_n} f(t) dt = \int_0^\infty f(t) dt = N$$

$$\lim_{n \rightarrow \infty} F(x_n) = N$$

$$\lim_{n \rightarrow \infty} \int_{x_n}^{y_n} f(t) dt = N - N = 0$$

